

A Nonstandard Analysis Approach to Toeplitz Operator Perturbations

TAYEB GUENFOUD

Preprint. Submitted for publication.

Abstract

The aim of this paper is to investigate perturbations of Toeplitz operators within the framework of nonstandard analysis and to derive new results.

1 Introduction

A Toeplitz operator is a fundamental object in operator theory and complex analysis. While many aspects of its stability are well understood, certain types of perturbations of a Toeplitz operator have not yet been studied. In this paper, we introduce and analyze a new class of perturbations using nonstandard analysis. This framework allows us to describe small and asymptotic effects in a precise way, leading to new results that are not accessible through classical methods. `amsmath`, `amssymb`, `amsthm` `mathrsfs` `hyperref`
`bm`

2 Nonstandard Analysis

Nonstandard analysis provides a rigorous framework for working with infinitesimal and infinite quantities. It allows us to study asymptotic behaviors and small perturbations in a precise way, which is particularly useful for analyzing perturbations of Toeplitz operators.

⁰Mathematics Subject Classification (2020): 47B35, 46S10, 46G10.

Keywords: Toeplitz operators, Bergman space, nonstandard analysis, operator perturbations.

2.1 Definitions

[Hyperreal Numbers] The set of *hyperreal numbers*, denoted by ${}^*\mathbb{R}$, extends the real numbers $\mathbb{R}$ by including infinitesimal and infinite numbers. Every real number $r \in \mathbb{R}$ has a corresponding hyperreal $r^* \in {}^*\mathbb{R}$.

[Infinitesimal] A hyperreal number $\epsilon \in {}^*\mathbb{R}$ is called *infinitesimal* if

$$|\epsilon| < \frac{1}{n} \quad \text{for all } n \in \mathbb{N}.$$

[Unlimited Number] A hyperreal number $H \in {}^*\mathbb{R}$ is *unlimited* (or infinite) if

$$|H| > n \quad \text{for all } n \in \mathbb{N}.$$

[Standard Part Function] For any finite hyperreal $x \in {}^*\mathbb{R}$, there exists a unique real number $\text{st}(x)$ such that

$$x \approx \text{st}(x),$$

where $\approx$ means x is infinitesimally close to $\text{st}(x)$.

[Internal Set] A set $A \subset {}^*\mathbb{R}$ is *internal* if it arises from the hyperextension of a standard set or from standard set operations within the nonstandard universe.

[S-Continuity / Infinitesimal Perturbation of Operators] A hyperreal operator T^* is an infinitesimal perturbation of a standard operator T if, for all standard vectors x ,

$$\|(T^* - T)x\| \text{ is infinitesimal.}$$

[Transfer Principle] Any first-order statement true for real numbers is also true for hyperreal numbers. This principle allows us to extend classical operator results to the nonstandard setting. For more detailed discussion, the reader is referred to [6].

3 Toeplitz Operators on the Bergman Space of the Unit Disc

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ denote the unit disc in the complex plane, and let $A^2(\mathbb{D})$ be the Bergman space, i.e., the Hilbert space of square-integrable holomorphic functions on $\mathbb{D}$ with the inner product

$$\langle f, g \rangle = \int_{\mathbb{D}} f(z) \overline{g(z)} dA(z),$$

where dA denotes the normalized area measure on $\mathbb{D}$.

[Toeplitz Operator] Let $\varphi \in L^\infty(\mathbb{D})$ be a bounded measurable function, called the *symbol*. The *Toeplitz operator* $T_\varphi : A^2(\mathbb{D}) \rightarrow A^2(\mathbb{D})$ is defined by

$$T_\varphi f = P(\varphi f),$$

where $P : L^2(\mathbb{D}) \rightarrow A^2(\mathbb{D})$ is the orthogonal projection onto the Bergman space.

Toeplitz operators on the Bergman space are bounded with $\|T_\varphi\| \leq \|\varphi\|_\infty$. They play a central role in operator theory, function theory, and complex analysis.

[Berezin Transform] The *Berezin transform* of a Toeplitz operator T_φ is the function $\tilde{\varphi} : \mathbb{D} \rightarrow \mathbb{C}$ defined by

$$\tilde{\varphi}(z) = \langle T_\varphi k_z, k_z \rangle,$$

where k_z is the normalized reproducing kernel at $z \in \mathbb{D}$.

The Berezin transform provides a bridge between the operator T_φ and its symbol φ , and is often useful in studying perturbations and spectral properties. For details and proofs, we refer the reader to [3].

3.1 Examples of Toeplitz Operators

- **Radial symbols:** $\varphi(z) = \phi(|z|)$, which often lead to diagonal operators with respect to the standard Bergman orthonormal basis.
- **Polynomial symbols:** $\varphi(z) = z^m \bar{z}^n$, useful for explicit computations and testing perturbation effects.

4 Nonstandard Perturbations of Toeplitz Operators on the Bergman Space

Using the framework of nonstandard analysis introduced in Section 2, we now define and study perturbations of Toeplitz operators on $A^2(\mathbb{D})$.

[Infinitesimal Perturbation of a Toeplitz Operator] Let T_φ be a Toeplitz operator on $A^2(\mathbb{D})$. A hyperreal operator T_φ^* is called an *infinitesimal perturbation* of T_φ if, for all standard $f \in A^2(\mathbb{D})$,

$$\|(T_\varphi^* - T_\varphi)f\| \text{ is infinitesimal.}$$

This definition formalizes the notion of a "small" or "negligible" perturbation in a precise nonstandard sense. Such perturbations can capture effects that are invisible to classical perturbation theory.

[Stability under Infinitesimal Perturbations] Let T_φ^* be an infinitesimal perturbation of T_φ . Then for all standard $f \in A^2(\mathbb{D})$, the standard part satisfies

$$\text{st}(\langle T_\varphi^* f, f \rangle) = \langle T_\varphi f, f \rangle.$$

By linearity and the definition of infinitesimal perturbation,

$$\langle T_\varphi^* f, f \rangle = \langle T_\varphi f, f \rangle + \langle (T_\varphi^* - T_\varphi) f, f \rangle,$$

where the second term is infinitesimal. Taking the standard part gives the result.

This result shows that infinitesimal perturbations do not change the "observable" values of Toeplitz operators on standard vectors, but they provide a framework to explore new asymptotic and limiting behaviors.

4.1 Example: Radial Symbol Perturbation

Let $\varphi(z) = |z|^2$ be a radial symbol and T_φ the corresponding Toeplitz operator. Consider a hyperreal perturbation $T_\varphi^* = T_\varphi + \epsilon T_\psi$, where ϵ is infinitesimal and $\psi \in L^\infty(\mathbb{D})$. Then T_φ^* is an infinitesimal perturbation of T_φ and can be analyzed using the framework above.

References

- [1] Albeverio, S., Fenstad, J. E., Høegh-Krohn, R., Lindstrøm, T., *Nonstandard Methods in Stochastic Analysis and Mathematical Physics*, Academic Press, 1986.
- [2] Axler, S. and Zhu, K., Toeplitz Operators on the Bergman Space, *Operator Theory: Advances and Applications*, Vol. 106, 1–20, 1998.
- [3] Berezin, F. A., Covariant and Contravariant Symbols of Operators, *Mathematics of the USSR-Izvestiya*, Vol. 8, 1107–1165, 1974.
- [4] Conway, J. B., *A Course in Functional Analysis*, Springer, 2000.
- [5] Douglas, R. G., *Banach Algebra Techniques in Operator Theory*, Springer, 1998.

- [6] Goldblatt, R., *Lectures on the Hyperreals: An Introduction to Nonstandard Analysis*, Springer, 1998.
- [7] Rudin, W., *Real and Complex Analysis*, McGraw-Hill, 1987.
- [8] Zhu, K., *Operator Theory in the Bergman Space*, Graduate Texts in Mathematics, Vol. 138, Springer-Verlag, New York, 2005.

Received: December 29, 2025

Email: tayeb.guenfoud@yahoo.com